

PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

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Abstract

001 ***Introduction.** Segmentation networks trained on one MRI*
002 *contrast or imaging modality often fail on another, and an-*
003 *notating data for every new domain is costly. Synthesis-*
004 *-based domain randomization such as SynthSeg generates*
005 *training images from label maps alone, but it needs dense*
006 *label maps and discards the texture of the real image. We*
007 *introduce PALETTE-Aug, a texture-preserving alternative*
008 *that needs only the task’s own labels, and ask when tex-*
009 *ture matters for generalization. **Methods.** PALETTE-Aug*
010 *remaps the real intensities of an unsupervised partition of*
011 *the image, so the texture inside every region survives. We*
012 *benchmark it against five methods on eight segmentation*
013 *tasks across MRI, CT and dental CBCT: five whose tar-*
014 *gets are bounded by an anatomical interface (healthy brain,*
015 *multi-organ, healthy spine, mandible, pelvis) and three de-*
016 *finied by tissue appearance (brain tumor, multiple sclero-*
017 *sis lesions, breast lesions). Each method trains on one*
018 *modality and is tested in-domain and out-of-domain. An*
019 *ablation over sixteen training settings holds the partition*
020 *fixed and replaces noise fill with the real-intensity remap.*
021 ***Results.** PALETTE-Aug wins the benchmark overall: it*
022 *has the highest Dice on seven of the eight tasks and, with*
023 *all tasks pooled, is significantly better than every competi-*
024 *tor. In the ablation, keeping texture raises Dice by +6.8*
025 *points on multiple sclerosis lesions and +4.6 on brain tu-*
026 *mor, both appearance-defined, against at most +3.5 on*
027 *interface-bounded tasks. Breast lesions, also appearance-*
028 *defined, gain only +1.5. **Conclusion.** Texture preservation*
029 *helps most on targets defined by tissue appearance rather*
030 *than an anatomical interface, though not on all of them.*
031 *PALETTE-Aug generalizes best overall across eight seg-*
032 *mentation tasks.*

033 1. Introduction

034 Medical image segmentation models are often trained on
035 one acquisition setting [18] and frequently applied to an-

other: a different scanner, sequence, or imaging modal-
ity than the one seen during training [17, 30]. MRI con-
trast is a protocol choice, not a physical constant, so the
same anatomy has different intensities across sequences [5].
A network trained only on T1-weighted brain images can
fail outright on T2-weighted or FLAIR images [3]. Ac-
quiring and annotating training data for every contrast a
model might encounter is not practical: manual segmen-
tation is expensive, and clinical sites rarely share raw data
across institutions [41, 60]. The setting that matters for de-
ployment is therefore single-source domain generalization:
train on one labeled dataset in one contrast, then generalize
to unseen contrasts and modalities without target-domain
data [45, 54, 65, 67].

The dominant single-source strategy is label-generative
domain randomization, popularized by SynthSeg [3].
SynthSeg does not use the real image at all: it builds each
training image from a label map, filling every labeled struc-
ture with Gaussian noise of a randomly drawn mean and
variance. The label map must be *dense*, a voxel-wise parcel-
lation of the whole image rather than the sparse annotation a
segmentation dataset normally provides. The resulting im-
ages have arbitrary, unrealistic contrast, forcing the network
to rely on shape and layout [3]. This is effective, but carries
two structural costs. First, it is *label-driven*: the dense par-
cellation it needs is richer than the task’s own annotation.
SynthSeg trains on whole-brain parcellations with dozens
of labeled structures [3], whereas an MS dataset such as
Open-MS annotates only the lesions [35]; obtaining the
parcellation reintroduces the expensive labeling the method
aims to avoid. Second, and less discussed, it *does not pre-*
serve texture: replacing every labeled region with piecewise
Gaussian noise discards the real gradients, partial-volume
boundaries, and unlabeled fine structure the acquired scan
contains. Layout survives; texture does not.

We ask when keeping texture, rather than discarding it,
helps contrast robustness. Our central finding is that the
answer depends on the *task*: preserving real intensity struc-
ture helps most where the target is defined by tissue appear-
ance, and much less where an anatomical interface sepa-

rates it from its surroundings. Prior work shows that texture carries diagnostic signal in appearance-defined tasks: texture features separate MS lesions from normal-appearing white matter [66], and diffuse glioma infiltrates beyond every margin visible on conventional MRI [20, 56].

PALETTE-Aug, our single-source augmentation, keeps this texture and needs no dense label map. **PALETTE-Aug** groups the foreground intensities of a scan into classes with 1-D k -means, splits each class into spatial sub-regions with a Voronoi partition, and applies a signed random affine remap to the *real* intensities of each sub-region. Edges, gradients and partial-volume detail therefore survive inside each sub-region, because existing intensities are remapped rather than replaced by noise. Only the contrast between regions is randomized, and without a target: **PALETTE-Aug** can produce any contrast, including implausible ones such as inverted intensities, because it is not tied to a specific target contrast. An optional step also re-randomizes each labeled structure; it uses whatever labels exist, a dense label map or just the task’s own segmentation labels.

We then test when texture matters, with a controlled ablation in sixteen training settings. We keep **PALETTE-Aug**’s partition and region target means and change only what fills each region: the region’s own image content, affinely remapped so that its gradients and fine structure survive (texture kept), or Gaussian noise (texture removed). The four largest Dice gains are all on appearance-defined targets: brain tumor (+12.7 points from T1c training, +6.0 from T1n) and MS lesions (+7.7 from FLAIR, +5.9 from T1w), against at most +5.9 in any interface-bounded setting. Pooled per task, the swap adds +6.8 Dice on MS and +4.6 on brain tumor, against at most +3.5 on an interface-bounded task. The converse does not hold: breast lesions, also appearance-defined, gain only +1.5, and several appearance-defined settings show no gain, so we state the result in one direction only (§4.3). A segmentation-free gradient measurement confirms that the swap changes texture: noise fill keeps only the partition edges, and real fill recovers most of the source texture.

We train one nnU-Net [27] configuration, varying only the epoch budget per dataset, and evaluate cross-contrast and cross-modality transfer on eight tasks: healthy brain, MS lesion, brain tumor, abdominal organ, mandible, breast lesion, spine and pelvis. We compare against nnU-Net with its default augmentation; the released SynthSeg (SynthSeg-noEM) and its expectation-maximization variant for cardiac MRI [3, §5.4] (SynthSeg-EM), whose code was not released, so we re-implemented it; SRCSM [57]; and the Auglab augmentation pipeline [42]. **PALETTE-Aug** wins the benchmark overall: it has the best Dice on seven of the eight tasks and significantly beats every competitor on five of them. Pooling all tasks into one test, it is significantly better than each competitor on both Dice and boundary dis-

tance, though not on every task.

In summary, we make the following contributions:

- **When texture preservation matters, tested by intervention.** Replacing noise fill with the real-intensity remap, with the partition fixed, gives its largest gains on appearance-defined targets, though not on every one. A segmentation-free gradient measurement confirms that the swap changes texture.
- **PALETTE-Aug**, a single-source contrast augmentation that needs no dense label map, only the task’s own labels. In every comparison it runs as one added operator inside the Auglab pipeline; the ablation also tests it alone.
- **Breadth evidence.** Across cross-contrast and cross-modality transfer on eight tasks and seventeen datasets, with one configuration reused throughout, **PALETTE-Aug** wins the benchmark overall: it is best on seven tasks and significantly beats every competitor on five; where it does not win, we test whether the leader is truly ahead.

2. Related Work

2.1. Domain generalization for medical segmentation

Domain generalization (DG) targets unseen distributions without access to target data, unlike unsupervised domain adaptation, which assumes unlabeled target images and solves an easier problem [53]. Most medical DG methods are *multi-source*, interpolating across several labeled domains by meta-learning, feature alignment, frequency-space exchange, multi-frequency architectures, foundation-model scale, or diffusion-based stylization [2, 36, 41, 44, 51, 60] – all of which still rely on source diversity. We instead work in the *single-source* regime (one labeled dataset, no target data, no second source), which is routine in clinical deployment [45, 67]. This rules out cross-domain interpolation and makes architecture-coupled methods orthogonal to our contribution: our mechanism acts at the input level and leaves the segmentation network a fixed nnU-Net [27].

2.2. Synthesis-based contrast-agnostic training

The dominant single-source strategy is label-generative domain randomization, introduced by SynthSeg [3]: intensities are sampled from a per-label Gaussian mixture over an anatomical segmentation, yielding images of arbitrary contrast that force contrast-invariance. This family has the two costs noted in Sec. 1: it needs a dense anatomical label map per subject, and it replaces real texture with piecewise noise. SIAM [58] trades label *quantity* for *quality* with curated templates, but still needs a full dense parcellation. The lineage has been specialized to white-matter and MS lesions and to atlas-based expectation-maximization [7, 8, 34]. Among SynthSeg’s own variants

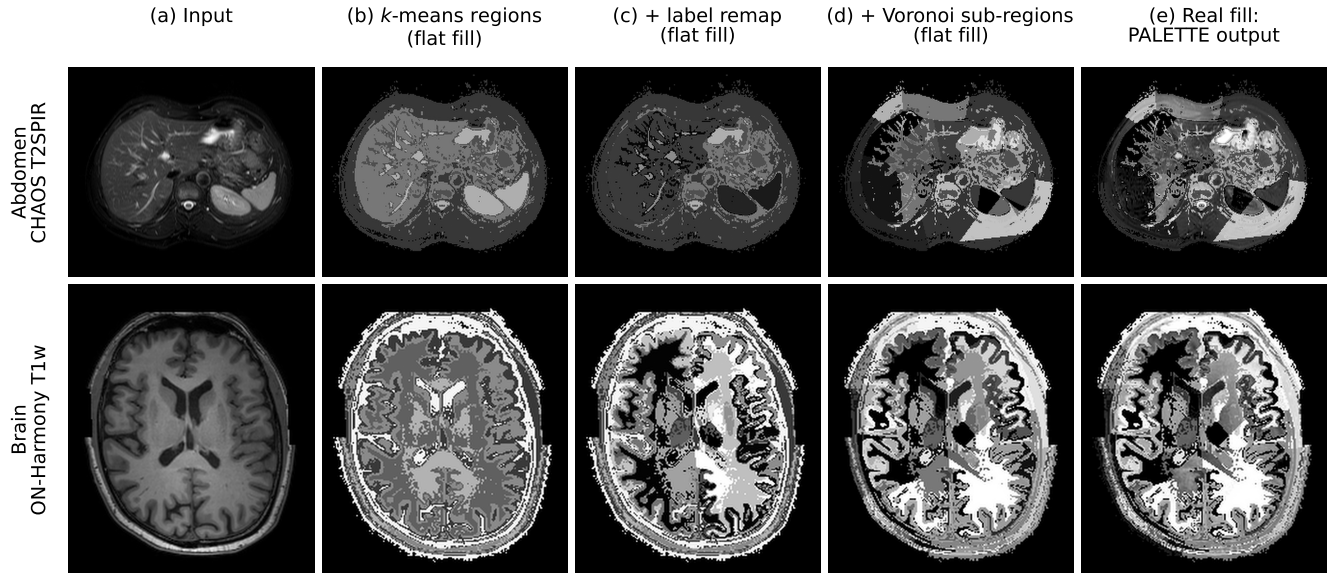


Figure 1. The PALETTE-Aug transform on one real CHAOS T2SPIR slice (top) and one real ON-Harmony T1w slice (bottom), built up in the order of its steps and of the ablation ladder (Secs. 3.1 and 4.3); every panel is a deterministic function of the same real image. (b)–(d) paint each region flat with its target mean μ (Eq. (1)) to show the partition alone: k -means regions (b, step 1); the label remap (c, step 2), an affine re-remap inside each anatomical label that keeps the structure already there (its straight edges are label boundaries, e.g. between the two hemispheres); Voronoi sub-regions under the same label remap (d, step 3). (e), the training input, fills the same partition through Eq. (1) from the real intensities (step 4), which keeps the texture inside every region.

we take the original (*noEM*), which synthesizes from the label map alone and never looks at the acquired scan, and the *EM* extension, which first splits every label, background included, into intensity-coherent sub-regions by expectation-maximization clustering of the real image, so that sparse task labels yield a richer map to synthesize from. SynthSeg introduced the EM extension for cardiac MRI [3, §5.4]; we re-implement it.

2.3. Texture as a segmentation signal

ImageNet-trained CNNs classify more by texture than by shape, and training them on style-randomized images shifts them toward shape and improves robustness [15]; how strong this texture bias is depends mostly on the training data and augmentation [23]; relying on either cue alone costs accuracy [40], and stylization’s robustness gain does not come from shape bias [43]. SynthSeg-style randomization pulls the same lever on medical images: replacing texture with noise forces the network onto shape. We ask what that costs when the target is itself defined by its appearance. In medical images, texture carries such information: second-order gray-level statistics discriminate image content beyond intensity [21], and texture features from tumor regions carry prognostic signal across CT, MRI and PET [33]. Kang et al. preserve texture in cross-domain translation with a co-occurrence loss, since distorted texture degrades downstream segmentation [29]. For diffuse

white-matter lesions, the lack of a sharp edge has been argued to favor texture-based over boundary-based segmentation [32]. None of this prior work, to our knowledge, connects texture preservation in an *augmentation* to *which* segmentation targets it helps; that connection, tested causally, is our contribution.

2.4. Single-source intensity and appearance augmentation

Where the first line of single-source methods synthesizes from a label map, a second manufactures appearance diversity directly from the real image, with no generative label model. General-purpose libraries stack intensity, spatial and artifact transforms on an nnU-Net trainer; the recent Auglab pipeline [27, 42] is one such GPU-accelerated framework and is the strongest competitor across our tasks, and BigAug stacks a comparable set of coarse intensity, quality and spatial perturbations under the same single-source DG protocol we use [67]. Closer to our operator are random-filter methods, which vary appearance with an untrained nonlinear filter: GIN applies randomly re-initialized convolutions for label-free appearance variants [45]; related variants establish the random-filter principle for natural images, perturb intensities per class using the task mask, add learned diffusion generators, or perturb the Fourier amplitude spectrum instead of the image itself [9, 37, 48, 54, 65, 68]. In natural images, style randomization rescales feature statistics,

keeping spatial structure [25, 39, 69], and Fourier methods change amplitude, keeping phase [64]. Our operator instead partitions the foreground into intensity classes and Voronoi sub-regions and applies a signed *affine* remap of the real intensities within each, invertible and gradient-preserving by construction. The closest peer is SRCSM [57]. Like PALETTE-Aug it perturbs region-wise in 3-D with one label-consuming step: it applies a fresh untrained convolution per anatomical class, smooths the resulting boundary artifacts, and histogram-matches test images to the source. They differ in the transform: SRCSM applies a random nonlinear convolution, PALETTE-Aug an affine remap that keeps the relative intensity structure inside each region.

3. Method

3.1. The PALETTE-Aug transform

Let x be a training volume min-max normalized to $[0, 1]$, x_v its intensity at voxel v , and $F = \{v : x_v > 0.01\}$ its foreground. PALETTE-Aug changes contrast region by region with one operation, a signed affine remap. We define it first, then build the regions it acts on in four steps, in the order in which the ablation adds them (§3.3, Fig. 1b–e).

Signed affine remap. For a region R , let $\bar{x}_R$ denote its mean intensity, $\mu_R \sim U(0, 1)$ a resampled target mean, and draw $\alpha_R \sim \pm U(0.5, 2.0)$ (a magnitude in $[0.5, 2.0]$ with an independently sampled sign). Every voxel $v \in R$ is remapped as

$$y_v = \mu_R + \alpha_R (x_v - \bar{x}_R). \quad (1)$$

Because Eq. (1) is affine in the *real* intensity x_v , it preserves the relative structure of the signal within the region: edges, gradients, and partial-volume transitions between adjacent voxels are scaled, not replaced. Only the contrast between regions is randomized, and $\alpha_R < 0$ also inverts local contrast.

1. Intensity partition. We run 1-D k -means on the foreground intensities $\{x_v : v \in F\}$ with k sampled per iteration from $\{2, \dots, 6\}$, producing k intensity classes $C_1, \dots, C_k$, each with its own target contrast (Fig. 1b). This partition groups voxels by intensity alone, with no label map, so it applies unchanged to any modality or contrast. Occasionally it is skipped and the whole foreground is remapped as one region; all step probabilities are in Sec. 7.

2. Anatomical-label remap (optional). An intensity partition does not always separate the regions of interest: where a target’s boundary is already faint, the partition can place both sides in one region, or give them similar contrasts, and weaken it further. Each anatomical label ℓ is therefore remapped on its own: we draw a new target mean

$\mu_\ell \sim U(0, 1)$ and signed scale $\alpha_\ell \sim \pm U(0.5, 2.0)$ and apply Eq. (1) *again* to ℓ ’s voxels after the region remap (Fig. 1c), so the label remap composes with the unsupervised one and keeps its internal structure. Since labels are independent of the partition, this restores a full-strength contrast step at every true label border; the anatomy itself is unchanged. It uses only the task’s own labels and is optional, *e.g.* on unlabeled data.

3. Spatial sub-parcellation. A Voronoi partition seeded with points sampled within each class C_i splits it into sub-regions $R_{i,1}, \dots, R_{i,m_i}$ that are spatially localized and similar in intensity, each with its own target contrast (Fig. 1d). Two voxels of equal intensity in different parts of the anatomy can thus fall in different sub-regions and receive independent contrast changes. Each class is usually, but not always, sub-split this way (Sec. 7).

4. Real fill. Every sub-region, and then every label, is filled through Eq. (1) from its own real intensities, which keeps the texture inside each region (Fig. 1e); the ablation replaces exactly this fill with noise (§3.3). Background voxels are zeroed and stay zero. The result is Gaussian-blurred half the time, before and optionally after the label remap, and a foreground z-score normalization gives the training input (Sec. 7).

3.2. PALETTE alone and PALETTE-Aug

We use the transform in two configurations, both applied on the fly on the GPU to half of the training iterations, with no learned parameters. *PALETTE alone* is the transform with spatial augmentation only (rotation, scaling, flips); it is also the real-fill rung of the ablation (§3.3). *PALETTE-Aug*, our full method, adds the transform as one operator to Auglab [42], a GPU nnU-Net augmentation pipeline [27] of intensity, spatial and artifact transforms and our strongest competitor (Sec. 4.2), so PALETTE-Aug and Auglab differ by exactly this transform.

3.3. Causal ablation: isolating texture preservation

PALETTE-Aug differs from SynthSeg in two ways at once: the partition (unsupervised k -means and Voronoi vs. an anatomical label map) and the fill (remapped real intensities vs. Gaussian noise). A direct comparison cannot separate them.

We construct a controlled variant, *noise-fill*, that keeps steps 1–3 of PALETTE-Aug (the same partition, label step and target mean μ_R for each region) and changes only step 4: instead of remapping each region’s real intensities with Eq. (1), it fills the region with SynthSeg-style Gaussian noise, $\mu_R + \sigma_R \epsilon$ with $\epsilon \sim \mathcal{N}(0, 1)$ i.i.d. per voxel and σ_R drawn uniformly from $[0.05, 0.25]$ per region. Labels are refilled the same way, in their own Voronoi cells, so both

Task	Target	Cohorts ([†] external, test only)	Trained on (one model each)	Tested on (every model)
Appearance-defined targets: abnormal signal inside a tissue				
Glioma	glioma, 4 sub-regions	BraTS2024-GLI [11]	T1n T1c T2w FLAIR	T1n, T1c, T2w, FLAIR
MS	MS lesions	Open-MS [35]	FLAIR T1w	FLAIR, T1w, T2w
Breast	tumor	I-SPY2 [38], Duke [†] [14, 50], I-SPY1 [†] [26], ACRIN-6698 [†] [46]	T1-CE T2w	T1-CE ^(†) , T2w, pre-contrast [†] , DWI [†]
Interface-bounded targets: a structure bounded by another tissue				
Abdomen	liver, kidneys, spleen	CHAOS [31], AMOS [†] [28], SLIVER07 [†] [22]	T1in T2spir	T1in, T1out, T2spir, T2w [†] , CT ^(†)
Brain	31 brain regions	ON-Harmony [61]	T1w T2w DWI	T1w, T2w, DWI, BOLD, EPI, GRE
Mandible	mandible	ToothFairy2 [6], HaN-Seg [†] [47], PDDCA [†] [49]	CBCT	CBCT, CT [†] , MR-T1 [†]
Spine	vertebrae, discs, canal	CT/Dixon cohort, SPIDER [†] [59]	CT Dixon in-phase	CT, Dixon in-phase, fat, water, T1w [†] , T2w [†]
Pelvis	11 bones and muscles	TotalSegmentator [10, 62]	CT MRI	CT, MRI

Table 1. The eight tasks and eighteen settings. Each training modality trains its own models, which are tested on every contrast listed for the task, the training contrast included ([†]: from an external cohort; ^(†): from the training cohort and an external cohort). All tasks but Spine have an ablation ladder. Sources and case counts per contrast: Tab. 5.

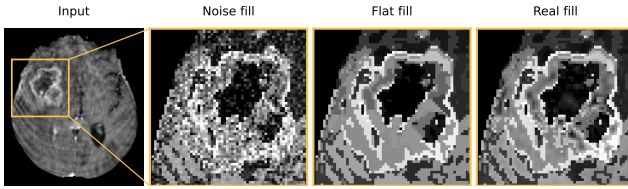


Figure 2. One partition and one set of remap draws (shared inside labels too, for illustration) on a glioma training slice (BraTS T1c), zoomed on the enhancing ring and filled three ways: Gaussian noise (the ablation’s noise fill), each region’s target mean (flat: the partition alone), or the remapped real intensities (PALETTE’s real fill).

arms have sub-regions inside labels. The partition and the target means are PALETTE-Aug’s own, so a difference between the two isolates whether the real intensity structure inside each region is kept or replaced by noise (Fig. 2).

We report this comparison separately for *interface-bounded* targets (a normal structure bounded by another tissue class) and *appearance-defined* targets (a pathology identified by abnormal signal within a tissue), assigned by anatomy; the texture hypothesis predicts a larger gain on the latter.

4. Experiments

4.1. Experimental setup

Tasks and data. Table 1 lists the eight tasks, their cohorts, the modalities each is trained on and the contrasts it is tested on, across MRI, CT and dental CBCT. Every (task, training modality) pair is one *setting*, eighteen in all; each setting trains its own models, which are tested on every contrast of the task, the training contrast included. *Out-of-domain*

(OOD) results exclude every scan of the training contrast, whichever cohort it comes from. External cohorts add held-out test contrasts to their task; they are not separate tasks.

Training and evaluation. Each model trains on one real source modality and is tested on every modality of its task, the protocol Auglab uses for spine [42], here over eight tasks. Splits are by subject: every scan of a test subject, in any modality, is held out of training, and no target-domain data is seen in training. All methods share the same 3D nnU-Net [27] configuration, optimizer, learning-rate schedule and per-task epoch budget (Sec. 7), and differ only in input augmentation. Each method trains three models on three train/validation folds, all scored on the same held-out test set. PALETTE’s configuration is set once and never tuned to a task (Sec. 7). Significance tests are paired over patients and Holm-corrected [24] (Sec. 9).

Compared methods. We compare against five methods (Tab. 2), each run as its authors propose. *nnU-Net* uses its default augmentation. *SynthSeg-noEM* is SynthSeg [3] as released. *SynthSeg-EM* is its EM label-subdivision variant, developed for cardiac MRI [3, §5.4]; its code was not released, so we re-implemented it. Both SynthSeg variants receive each task’s training labels; SynthSeg-noEM synthesizes from the label map alone and was designed for dense labels, which only Brain provides. *SRCSM* [57] and *Auglab* [42] use their default configurations (Tab. 4). SynthSeg and SRCSM run without the Auglab pipeline, as published (Sec. 5).

Method	Intensities	Varied per	No dense labels	Real texture
nnU-Net	real	—	✓	✓
SynthSeg-noEM	Gaussian	label	×	×
SynthSeg-EM	Gaussian	EM sub-label	✓	×
SRCSM	random convolution	label	✓	~
Auglab	real, generic aug.	image	✓	✓
PALETTE-Aug	real, affine remap	sub-region	✓	✓

Table 2. What each method trains on. Sub-region: unsupervised k-means and Voronoi partition (Sec. 3). No dense labels: works from the task’s own labels (SynthSeg-noEM runs on them but fills everything unlabeled with one Gaussian). ~: the real image passes through a random filter. Augmentation probabilities: Tab. 4.

4.2. Main comparison

PALETTE-Aug is the best method overall. PALETTE-Aug has the best Dice on seven of the eight tasks (Tab. 3). We test its advantage at two levels. Within each individual task (bottom row), it significantly outperforms every competitor on five tasks (Glioma, MS, Breast, Spine, Pelvis), all but SRCSM on Abdomen (85.6 vs. 85.9), and three of five on Brain and Mandible. Pooling the test patients of all eight tasks, each task weighted equally, it is significantly better than every competitor: $p = 1.1 \times 10^{-13}$ against Auglab, the closest, and $p \leq 10^{-35}$ against the others. With patients as the unit, this pooled test generalizes to new patients of these eight tasks, not to new tasks; with the task as the unit, PALETTE-Aug beats Auglab on seven of eight tasks (one-sided sign test $p = 0.035$) and SRCSM on six of the seven tasks where SRCSM was run ($p = 0.063$).

SynthSeg-noEM falls below nnU-Net on the six sparse-label tasks; on Brain, the one dense-label task, it is within 2.6 Dice of PALETTE-Aug.

No competitor is reliably second. The best competitor depends on the task: Auglab on Glioma, MS, Breast, Spine and Pelvis, SRCSM on Abdomen, Brain and Mandible. SRCSM is also the weakest method on Breast (14.7 Dice). Picking the right competitor would require knowing the answer in advance; PALETTE-Aug is first on every task but Mandible.

Where PALETTE-Aug does not win. On Mandible, SRCSM is first (79.8), Auglab second (78.3) and PALETTE-Aug third (77.7). The one-sided test above cannot show that a competitor is ahead, so we also ran the reverse test (same design, Holm-corrected across competitors within the task): only Auglab is significantly ahead of PALETTE-Aug (+0.6 Dice, $p = 3 \times 10^{-3}$); SRCSM’s +2.1 is not significant ($p = 0.15$). Interestingly, PALETTE alone ties Auglab (78.3).

PALETTE without Auglab. PALETTE alone (Sec. 3.2) is as strong as the full method: over the seven tasks with both, it is significantly better than every competitor, Auglab included (Dice $p = 4 \times 10^{-5}$), and does not differ significantly from PALETTE-Aug (two-sided $p = 0.60$). PALETTE-Aug’s margin over the competitors therefore does not come from the Auglab pipeline. Per task, adding Auglab raises Dice on Glioma and Breast, lowers it on MS (significantly) and slightly on Abdomen, Brain and Mandible, and leaves Pelvis unchanged (Tab. 3). PALETTE thus composes with a full augmentation pipeline without losing its advantage, but which operators help alongside it depends on the task, and choosing them per task may add more; we have not identified which Auglab operator is responsible for the losses.

Boundary distance. On HD95 (Tab. 9), PALETTE-Aug is lowest on six of the eight tasks and significantly better than every competitor with all tasks pooled; it ties Auglab on Breast and trails Auglab and nnU-Net on Mandible.

4.3. Causal ablation of texture preservation

Does keeping the real texture help beyond PALETTE-Aug’s partition, and on which targets? We test this with a ladder: nnU-Net $\rightarrow$ +k-means $\rightarrow$ +label remap $\rightarrow$ +Voronoi $\rightarrow$ real-intensity fill $\rightarrow$ +Auglab. The first rung is nnU-Net with its default augmentation, as in Tab. 3. The next four rungs share one pipeline (Sec. 3), and each adds one ingredient. The three partition rungs fill every region with SynthSeg-style Gaussian noise. The step from +Voronoi to real fill, the *fill swap*, changes only the fill: partition, sub-region count, target-mean distribution and both blur passes are unchanged (inside labels, the noise arm draws its own Voronoi cells). Any change at this step therefore comes from keeping the real within-region intensity structure instead of filling the region with noise, the same Gaussian noise as SynthSeg.

In our data every large gain from real fill is on an appearance-defined target, but not every appearance-defined target shows one (Fig. 3 and Tab. 10). The swap is tested once per task, pooling its training modalities (Sec. 9); the per-setting values are a breakdown of that test. Pooled per task, the swap adds +6.8 Dice on MS (all eight test patients improve) and +4.6 on Glioma, against +3.5 on Abdomen, +3.1 on Pelvis, −0.7 on Brain and −1.6 on Mandible; each of these tasks passes the correction except Mandible, Brain with a loss. The four largest per-setting gains are Glioma T1c (+12.7 Dice), MS FLAIR (+7.7), Glioma T1n (+6.0) and MS T1w (+5.9); no interface-bounded setting gains more than +5.9 (Pelvis CT). Breast gains only +1.5, mostly from T2w training (+2.7; T1-CE training +0.4), and Glioma T2w and FLAIR show no gain (+1.1, −1.3). Relative to the noise-fill Dice (Fig. 3) the

	Glioma	MS	Breast	Abdomen	Brain	Mandible	Spine	Pelvis	p vs ours
nnU-Net	23.4	21.0	35.0	29.5	34.5	59.2	48.5	29.8	3×10^{-75}
SynthSeg-noEM	8.8	4.3	16.5	60.9	64.1	53.6	47.5	15.0	2×10^{-86}
SynthSeg-EM	43.1	33.9	39.6	84.0	66.1	76.3	80.6	62.5	1×10^{-35}
SRCSM	36.9	27.5	14.7	85.6	66.5	79.8	—	62.9	8×10^{-66}
Auglab	44.8	34.3	40.1	84.6	66.0	78.3	80.9	64.3	1.1×10^{-13}
PALETTE alone (ours)	43.7	41.1	40.3	86.4	67.0	78.3	—	65.7	0.70 [§]
PALETTE-Aug (ours)	45.9	38.5	41.2	85.9	66.7	77.7	82.3	65.6	—
competitors beaten ($p < 0.05$)	5/5	5/5	5/5	4/5	3/5	3/5	4/4	5/5	

Table 3. Task-level Dice (%). Each column pools the task’s test cases of every contrast (the training contrast included), all source modalities and any external cohorts, weighting contrast groups equally so a modality with several external sources does not outvote one with fewer (Sec. 9). p : Holm-corrected one-sided sign-flip test on the macro-averaged per-patient difference, stratified by task with tasks weighted equally. We print no cross-task mean Dice, since whole-liver and breast-lesion Dice are not on a common scale. The bottom row counts the competitors PALETTE-Aug beats within each task alone (same test). SRCSM includes its published test-time source matching. Spine has no SRCSM run (Tab. 5). PALETTE alone: the PALETTE transform without the Auglab pipeline (the ladder’s real-fill rung, Sec. 4.3), left out of the bolding; Spine has no ladder. [§]Same test, PALETTE-Aug against PALETTE alone, over the seven tasks with both. HD95 is in Tab. 9.

ranking at the top is unchanged.

ON-Harmony labels 31 brain regions in the FreeSurfer label set [13]. Nearly all are bounded by another tissue class (CSF against tissue, grey against white matter), so we group it with Abdomen as interface-bounded. At the task level it is the only task where the swap significantly hurts (-0.7 Dice), with sessions as units. It is also the only task labeling nearly the whole brain; with one such task, we cannot tell whether that is why.

The fill swap changes texture. The ladder shows that the fill swap changes Dice, not that it changes texture. We check this without any segmentation, with the normalized-gradient-field (NGF) similarity [19] between each augmented volume and its source scan. NGF measures how well image gradients align: it is 1 when all edges and fine structure are kept and about $1/3$ for isotropic noise, the mean squared cosine between independent random 3-D directions. Figure 4 reports it for every method on the seven ablation tasks. The noise fill keeps only the partition edges (mean NGF 0.44); the real fill, on the same partition, raises it to 0.83, the highest of all methods. PALETTE alone keeps more texture than every competitor on six tasks and ties with Auglab on Brain (Tab. 18). SynthSeg keeps little (0.50 with EM, 0.37 without, near the noise floor). PALETTE-Aug keeps less (0.70) than both PALETTE alone and Auglab (0.74): Auglab’s operators act on the remapped image, and together they remove more texture than either alone. It still keeps more than SRCSM (0.64) and both SynthSeg variants.

Supporting analyses. The supplement gives per-setting results (Sec. 10), every ladder rung, every training $\rightarrow$ test

pair and the HD95 ablation (Sec. 11), and a component ablation on all test contrasts (Sec. 13).

5. Discussion and Limitations

A possible reason for the largest effects. Liver, spleen and kidney are encapsulated organs with a clear physical boundary, and edge-following methods have long segmented them well [55]. An augmentation that scrambles their internal texture but keeps the boundary loses little. Diffuse glioma has no such boundary: tumor cells infiltrate beyond both the enhancing margin and the FLAIR abnormality [20, 56], so the target must be inferred from appearance. MS lesions, though typically well demarcated, are identified by abnormal signal within white matter, and texture features separate them from normal-appearing white matter [66]. This account interprets the intervention of Sec. 4.3; it is not independent evidence.

Not every appearance-defined target gains much. Glioma T2w and FLAIR show no Dice gain, and Breast gains only slightly: $+2.7$ Dice from T2w training and $+0.4$ from T1-CE. We have two hypotheses. First, the effect depends sharply on the training contrast (Glioma: $+12.7$ from T1c, $+6.0$ from T1n, $+1.1$ and -1.3 from T2w and FLAIR), so whether its texture transfers may matter as much as the target; with T2w training, real fill helps some labels and hurts others (Tab. 15). Second, Breast pools four cohorts from different trials, which may dilute a cohort-specific effect. On interface-bounded tasks, where texture adds little relative to the starting Dice, most of the gain arrives with the texture-free partition rungs (Tab. 12), that is, from the wide range of region-to-region contrasts PALETTE-Aug samples.

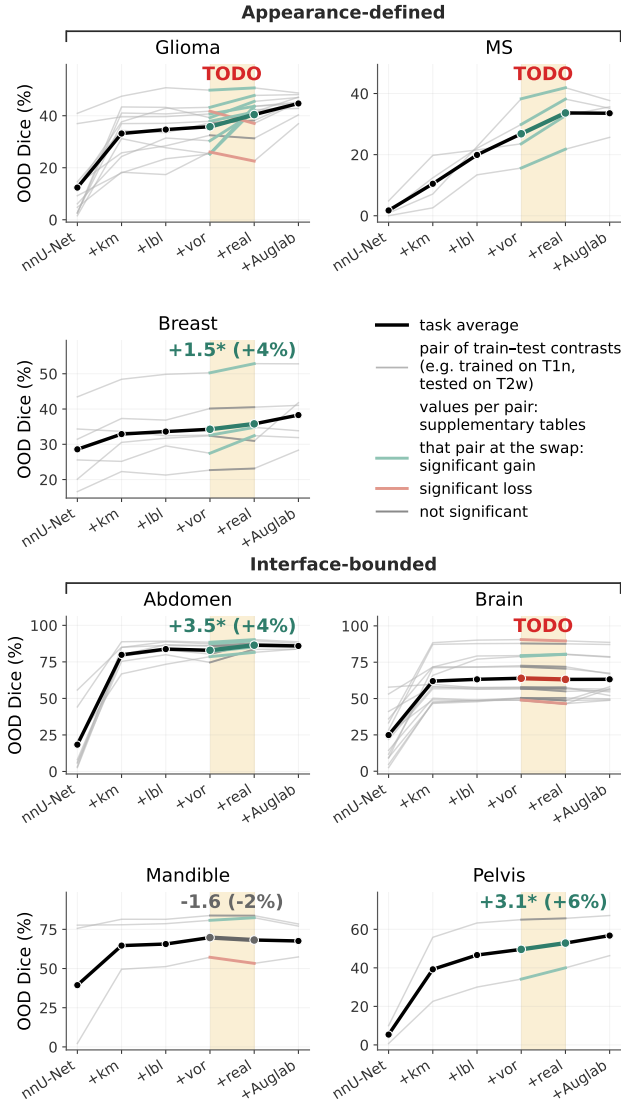


Figure 3. The fill swap (shaded band), one panel per task with a ladder, grouped by target type. Thin lines: one model, i.e. one training modality tested on one held-out contrast (external cohorts included), colored at the swap by that pair’s significance; values per pair: Tab. 13. Bold line: the task average. Annotation: the task’s Dice change at the swap in points, and relative to its noise-fill Dice in brackets; *: significant (paired Wilcoxon on patients pooling all training modalities, Holm-corrected across the seven tasks: the one correction family of the fill swap; Brain’s eight sessions come from four subjects). Per setting: Tab. 10; HD95: Fig. 5.

Limitations. We assign each target to a group by its anatomy, not by a measured boundary statistic, and the grouping is ours. Target type is confounded with difficulty: all interface-bounded rows but one start above 63% noise-fill Dice and every appearance-defined row below 42%. We therefore lead with the absolute Dice change; dividing by the starting Dice favors the appearance-defined rows, and

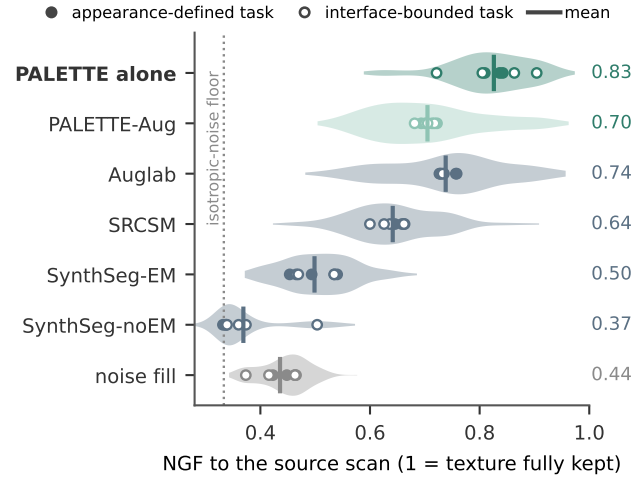


Figure 4. Texture kept by each augmentation: NGF similarity of the augmented volume to its source scan on the seven ablation tasks (violin: all scans, each averaged over 5 draws; dots: task means; bar: mean of the task means). The noise fill keeps only the partition edges; the real fill keeps the most.

as a share of the remaining error (100% minus noise-fill Dice) Abdomen T2spir would rank first. The exception, Pelvis CT, starts at 34% and gains +5.9, below the four large appearance-defined gains. The between-group comparison is descriptive: a permutation test on the seven per-task gains gives $p = 0.2$. MS has eight test patients and Brain eight sessions of four subjects, the floor of a paired test, and Brain’s session-level tests are optimistic. Spine was run on a separate pipeline and has no SRCSM arm. The cross-task test covers new patients, not new tasks; with the task as the unit, the advantage over SRCSM is not significant. We applied each competitor as its paper proposes and did not place SRCSM or SynthSeg inside the Auglab pipeline; PALETTE alone, without Auglab, already beats each of them (Sec. 4.2).

6. Conclusion

Preserving real image texture helps most where the target is defined by tissue appearance (brain tumor, MS lesions: +4.6 and +6.8 Dice per task) rather than by an anatomical interface (at most +3.5), though not on every appearance-defined target. PALETTE-Aug, which preserves texture, needs no dense label map and was never tuned to a task, wins the eight-task benchmark overall. All code is public (link withheld for review), as are all datasets but the Spine training cohort.

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PALETTE-Aug: A Simple Texture-Preserving Augmentation for Medical Imaging Domain Generalization

Supplementary Material

884 7. Training configuration

885 All methods share one nnU-Net [27] configuration; only the
886 input augmentation differs. Nothing below is tuned per task.

887 **Training schedule.** 3D full-resolution nnU-Net, SGD
888 with Nesterov momentum 0.99, initial learning rate 10^{-2}
889 with the nnU-Net `polyLR` decay, 250 mini-batches per
890 epoch, combined Dice and cross-entropy loss, deep supervision on. Patch size, spacing and batch size are whatever the
891 nnU-Net planner selects for that dataset, identically for every
892 method. The epoch budget is fixed per task and shared
893 by all methods on it: 2500 (Glioma), 2000 (Brain, MS),
894 1000 (Mandible, Breast) and 200 (Abdomen, Pelvis); Spine
895 was trained on a separate pipeline. Each method is trained
896 as three models on three train/validation folds, all scored on
897 the same held-out test set; folds are the same across methods.
898
899

900 **PALETTE-Aug hyper-parameters.** Set once and never
901 tuned per dataset: k -means class count drawn per sample
902 from $\{2, \dots, 6\}$, skipped in favor of a single global
903 remap with probability 0.10; Voronoi sub-parcellation of
904 each intensity class with probability 0.60 (else left as one
905 region); signed affine coefficient $\alpha \in \pm U(0.5, 2.0)$ applied
906 per sub-region as in Eq. (1) (always applied, to whatever
907 partition results from the k -means/Voronoi steps); per-label
908 affine decoupling applied independently per label with
909 probability 0.5 (skipped for any label with under 4 foreground
910 voxels in that sample); Gaussian blur σ drawn from
911 $\{0, 0, 0, 0.3, 0.5, 0.8\}$, applied twice (after the region-level
912 and after the label-level remap), so applied half the time at
913 each point. The transform is applied on the GPU, on-the-fly,
914 to a fixed 50% of training iterations; the remaining half
915 of iterations use the un-transformed real image.

916 **Validation-time configuration.** Every reported model,
917 every competitor and every ladder rung included, selects its
918 best checkpoint on real-contrast validation images (*val000*).
919 *val100* applies PALETTE-Aug to the entire validation split
920 instead; it is reported only in Sec. 12.

921 **Competitor configurations.** Both SynthSeg variants run
922 through Auglab's GPU implementation of the SynthSeg
923 generator, with uniform intensity priors and with the generator's
924 own spatial and flip steps disabled (the pipeline's spatial
925 augmentation is shared by all methods). They receive

each task's training label map; SynthSeg-EM additionally
subdivides it on the fly by expectation-maximization clustering
of the image (2 sub-regions per foreground label, 3–10 for the
background, 20 iterations), where the original work precomputes
these maps [3, §5.4]. Each sub-region then receives a randomly
sampled intensity, as in SynthSeg. Both SynthSeg variants and
SRCSM synthesize on every training iteration because their original
papers train that way, so we keep that setting when running them;
PALETTE-Aug synthesizes on 50% of iterations (Tab. 4 lists every
augmentation's probability). SRCSM is run as published, including
its test-time source matching: each test image is histogram-matched
to the average intensity histogram of the training images (re-implemented
from the authors' code), with two adjustments for our data: MR
intensities are clipped at 0 first (resampling leaves small negative
values that would otherwise map the background to a tissue intensity)
and the CT window is $[-1024, 3071]$ instead of $[-1023, 1024]$ (CBCT
and bone exceed 1024). Held-out cases are never part of the source
histogram. Every method is predicted with the same nnU-Net predictor,
without test-time augmentation. SRCSM and Auglab otherwise use
their published default configurations. No competing method was
re-tuned in our favor, and all use the same folds, schedule and epoch
budget as above.

Compute. Training was performed on NVIDIA L40S and H100 GPUs.
A single fold costs from under an hour (Abdomen, 200 epochs) to
roughly three days (Glioma, 2500 epochs); the full study is dominated
by the six methods $\times$ sixteen settings $\times$ three folds of the main
comparison (Spine's two settings were trained on a separate pipeline,
without SRCSM), plus the sixteen ablation ladders of Sec. 4.3.

	Transform	nnU-Net	SynthSeg-noEM	SynthSeg-EM	SRCSM	Auglab	PALETTE alone	PALETTE-Aug
Synthesis	SynthSeg generator	–	1.0	1.0	–	–	–	–
	semantic random convolution	–	–	–	1.0	–	–	–
	PALETTE transform	–	–	–	–	–	0.5	0.5
Intensity	Gaussian noise	0.1	0.1	0.1	–	0.1	–	0.1
	Gaussian blur	0.2	0.2	0.2	–	0.2	–	0.2
	brightness	0.15	0.15	0.15	–	0.15	–	0.15
	contrast	0.15	0.15	0.15	–	0.15	–	0.15
	gamma / inverted gamma	0.3 / 0.1	0.3 / 0.1	0.3 / 0.1	–	0.3 / 0.1	–	0.3 / 0.1
	simulated low resolution	0.25	0.25	0.25	–	0.25	–	0.25
	bias field	–	0.15	0.15	–	0.15	–	0.15
	random intensity function	–	0.1	0.1	–	0.1	–	0.1
	Scharr / unsharp mask	–	–	–	–	0.25 / 0.25	–	0.25 / 0.25
	random convolution	–	–	–	–	0.2	–	0.2
	label-wise intensity redistribution	–	–	–	–	0.4	–	0.4
	intensity inversion	–	–	–	–	0.25	–	0.25
	histogram equalization	–	–	–	–	0.25	–	0.25
Spatial	rotation / scaling	0.2 / 0.2	0.2 / 0.2	0.2 / 0.2	–	0.2 / 0.2	0.2 / 0.2	0.2 / 0.2
	affine + elastic deformation	–	–	–	1.0	–	–	–
	flip	0.5 ^a	0.5	0.5	–	0.5	0.5	0.5

Table 4. Per-iteration probability of each augmentation, per method (–: not applied). The synthesis rows are each method’s own contrast step: the SynthSeg variants and SRCSM synthesize on every iteration, as in their original papers, PALETTE alone and PALETTE-Aug on half of them. PALETTE alone (the ladder’s real-fill rung) has no intensity augmentation besides the PALETTE transform. The remaining rows are the method’s generic augmentation pipeline: nnU-Net’s defaults for nnU-Net, Auglab’s for the Auglab-based methods (the SynthSeg variants switch off Auglab’s appearance-randomizing operators, which would otherwise act on top of the synthetic image), and SRCSM’s own spatial recipe. ^annU-Net mirroring, each axis independently.

8. Datasets and evaluation protocol

All evaluation is performed on held-out test cases only. Each model is trained on one modality of its task and evaluated on every test contrast of that task, the training contrast included; training and validation subjects never appear in any reported number.

Glioma (BraTS2024-GLI [11]). Post-treatment glioma sub-region segmentation (necrotic core, surrounding non-enhancing FLAIR hyperintensity (SNFH), enhancing tumor, resection cavity), trained on each of T1n, T1c, T2w and FLAIR and evaluated on all four.

MS (Open-MS [35]). MS-lesion segmentation with manual consensus annotations, evaluated across FLAIR, T1w and T2w.

Breast (I-SPY2; external: Duke, I-SPY1, ACRIN-6698 [14, 26, 38, 46, 50]). Breast-lesion segmentation on malignant-only cohorts. I-SPY2 trains, on its first-post-contrast T1 (T1-CE) and its T2w; its reference is the trial’s functional tumor volume mask. Three external cohorts are test-only. Duke and I-SPY1 come through the MAMA-MIA release, whose expert primary-lesion protocol both share, and contribute T1-CE and pre-contrast columns. ACRIN-6698, the diffusion sub-study of I-SPY2 with no patient in

common with our I-SPY2 subjects, contributes the baseline diffusion-weighted acquisition with the trial’s whole-tumor DWI region, a contrast neither training modality resembles. Because I-SPY2 is mostly acquired unilaterally, bilateral external volumes are cropped to the lesion side and, for the full-thorax acquisitions, also anterior-posteriorly to the breast-plus-chest-wall extent of I-SPY2’s own crops; without the second crop every method’s HD95 is dominated by the thorax. I-SPY2’s DCE series are acquired unilaterally at some sites and bilaterally at others; both field-of-view variants are kept per patient, and the significance tests treat them as one patient.

Abdomen (CHAOS [31]; external: AMOS [28], SLIVER07 [22]). Abdominal organ segmentation (liver, spleen, both kidneys), evaluated on the CHAOS MR contrasts (four held-out patients) and CT (20 held-out patients), and on two external cohorts: CT (100) and T2w MRI (20) from AMOS and liver-only CT (20) from SLIVER07. In the pooled CT group, liver comes from CHAOS, AMOS and SLIVER07 and spleen and kidneys from AMOS only (CHAOS CT is annotated for the liver only). The full-torso external volumes are cropped to the axial field of view of CHAOS before prediction (field-of-view matching, below).

Brain (ON-Harmony [61]). 31-region healthy-brain segmentation. Reference labels are *silver standard*: FreeSurfer

Task	Target	Trained on (one modality at a time)	Other test contrasts	External test cohorts	Test cases	Ladder
<i>Appearance-defined</i>						
Glioma	glioma, 4 sub-regions	BraTS2024-GLI [11]: T1n, T1c, T2w, FLAIR	–	–	70	✓
MS	MS lesions	Open-MS [35]: FLAIR, T1w	T2w	–	8	✓
Breast	tumor	I-SPY2 [38]: T1-CE, T2w	–	Duke [14, 50], I-SPY1 [26]: T1-CE, pre-contrast; ACRIN-6698 [46]: DWI	84 + 829	✓
<i>Interface-bounded</i>						
Abdomen	liver, kidneys, spleen	CHAOS [31]: T1in, T2spir	T1out, CT	AMOS [28]: CT, T2w; SLIVER07 [22]: CT	24 + 140	✓
Brain	31 brain regions	ON-Harmony [61]: T1w, T2w, DWI	BOLD, EPI, GRE	–	8 [†]	✓
Mandible	mandible	ToothFairy2 [6]: CBCT	–	HaN-Seg [47]: CT, MR-T1; PDDCA [49]: CT	71 + 82	✓
Spine	vertebrae, discs, canal	CT/Dixon cohort: CT, Dixon in-phase	Dixon fat, water	SPIDER [59]: T1w, T2w	29 + 171/219	–
Pelvis	11 bones and muscles	TotalSegmentator [10, 62]: CT, MRI [‡]	–	–	30/27	✓

Table 5. Per-task dataset roster (summary in Tab. 1, main paper). Test cases: own held-out split + external cohorts; “ x/y ” gives the counts of two contrasts where they differ (per dataset and contrast: Tab. 6). Ladder: the task has the causal ablation of Sec. 4.3. [†]Sessions of four subjects. [‡]Different patients for CT and MRI.

1007 parcellations [13] of each session’s T1w, grouped into 31 re-
 1008 gions and rigidly registered with ANTs into the native space
 1009 of the same session’s other contrasts. They are not manual
 1010 annotations, so absolute Dice here is agreement with an au-
 1011 tomated reference; every method is scored against the same
 1012 reference. The test set is eight sessions of four subjects,
 1013 each scanned at two sites.

1014 **Mandible (ToothFairy2; external: HaN-Seg, PDDCA [6,**
 1015 **47, 49]).** Mandible segmentation, trained on dental cone-
 1016 beam CT, the one non-CT, non-MRI training modality here.
 1017 ToothFairy2’s annotation coverage is cohort-dependent, so
 1018 every column of this task is scored on the mandible alone
 1019 (teeth excluded, matching the external references), which
 1020 keeps the numbers being averaged commensurable. Tooth-
 1021 Fairy2 acquires CBCT only, so apart from its own 71 CBCT
 1022 test cases its held-out contrasts all come from other datasets:
 1023 HaN-Seg CT (42 cases) and MR-T1 (41) and PDDCA CT
 1024 (40), the two CT sources pooled as one contrast group.
 1025 ToothFairy2’s volumes are stored superior-inferior inverted
 1026 relative to the external cohorts — an inversion invisible to
 1027 the image header, found by checking anatomy directly —
 1028 so the external test volumes are flipped into ToothFairy2’s
 1029 convention before inference, which leaves Dice and HD95
 1030 unchanged under the rigid transform. The correction mat-
 1031 ters for the comparison: a method without flip augmentation
 1032 (SRCSM) was penalized by the inversion far more than the
 1033 others, and it is after this correction that SRCSM leads this
 1034 task.

Spine (CT/Dixon cohort; external: SPIDER [59]). Ver-
 tebrae, intervertebral discs and spinal canal. Results for this
 task were computed on a separate pipeline with the same
 method suite except SRCSM, trained on CT and Dixon in-
 phase MRI of a CT/Dixon cohort and tested on that cohort’s
 held-out CT and Dixon fat, in-phase and water images and
 on the T1w and T2w of SPIDER. Dice and HD95 are com-
 puted on each whole structure per label. No SRCSM arm
 was run, so Spine is absent from every SRCSM comparison.

Pelvis (TotalSegmentator [10, 62]). Eleven pelvic and
 hip musculoskeletal structures (both hips, sacrum, and the
 gluteus maximus, medius and minimus and iliopsoas on
 each side), from the TotalSegmentator CT and MRI re-
 leases. The two are unpaired cohorts, and the MRI re-
 lease mixes sequences by design, so its MRI column is a
 sequence-heterogeneous test rather than one named con-
 trast. CT is subsampled to 200 cases to keep training time
 comparable with the other tasks.

Field-of-view matching. A model tested on a field of
 view it never saw in training can fail for reasons that have
 nothing to do with contrast or modality. Where an exter-
 nal cohort images more of the body than the training co-
 hort, we therefore crop its volumes before prediction, so
 that the shift left between training and test is, as far as pos-
 sible, the contrast or modality alone. Abdomen (AMOS,
 SLIVER07): the axial slab CHAOS covers, placed with the
 median CHAOS margins around the kidneys (AMOS) or
 the liver (SLIVER07). Mandible (HaN-Seg, PDDCA): a
 144 × 128 × 80 mm box centered on the mandible, close

to the 90th percentile of ToothFairy2’s training extent, enlarged (at most to $154 \times 154 \times 89$ mm) only when needed to contain the whole mandible. Breast (Duke, ACRIN-6698): the lesion-side half of bilateral volumes, then, for full-thorax acquisitions, an anterior-posterior window from 27 mm in front of the skin to 147 mm behind it, the measured geometry of I-SPY2’s unilateral acquisitions, extended if needed to contain the lesion; I-SPY1 is natively unilateral and is not cropped. The network sees, and is scored on, only the cropped volume, and the crops are identical for every method.

Task	Dataset (role)	Train	Test cases
Glioma	BraTS2024-GLI (train)	630	70
MS	Open-MS (train)	22	8
Breast	I-SPY2 (train)	476	84 patients ^a
	Duke (external)	–	291
	I-SPY1 (external)	–	167 ^b
	ACRIN-6698 (external)	–	371
Abdomen	CHAOS (train)	16	4 MR, 20 CT
	AMOS (external)	–	100 CT, 20 T2w
	SLIVER07 (external)	–	20 CT
Brain	ON-Harmony (train)	84 ^c	8 sessions
Mandible	ToothFairy2 (train)	408	71
	HaN-Seg (external)	–	42 CT, 41 MR-T1
	PDDCA (external)	–	40 CT
Spine	CT/Dixon cohort (train)	n/a ^d	27–29 per contrast
	SPIDER (external)	–	171 T1w, 219 T2w
Pelvis	TotalSegmentator CT (train)	170	30
	TotalSegmentator MRI (train)	156	27

Table 6. Every dataset, by task. Train: subjects in the three train/validation folds (identical across folds and methods); external cohorts are test-only. Test cases per contrast where they differ. ON-Harmony counts are scan sessions (the eight test sessions are four subjects, each scanned at two sites). ^aOne held-out patient can yield two T2w images (both field-of-view variants). ^bAfter excluding three bilateral cancers and one implant by a rule fixed before prediction. ^c76 for the T2w-trained models. ^dTrained on a separate pipeline. Pelvis MRI keeps 183 of the 298 released cases.

9. Statistical testing

Test construction. All tests are paired, and the unit of independence is the *patient*, never the fold, the image or the contrast. A patient’s score is its mean over labels and evaluated folds; where the same patient is scored in several held-out contrasts, or by the models of several training modalities, those scores move together as one unit under the null. Treating them as separate samples would be pseudo-replication and would inflate the effective sample size — a real risk here, since several tasks test a handful of patients under many contrasts (Brain: eight sessions of four subjects, six contrasts). For Brain the unit is the session, since its four subjects leave a test that could never reach significance; two sessions of one subject are not fully independent, so its *p*-values are, if anything, optimistic, and we

state Brain-only results with this caveat. Per-setting comparisons (Sec. 10) use the same test within each setting. The ablation ladders use the paired Wilcoxon signed-rank test [63], since there each task is judged on its own: each patient contributes one difference, its mean over the held-out contrasts and training modalities it appears in, so patients, not contrasts, are weighted equally. The fill swap has one correction family: it is tested once per task and Holm-corrected across the seven tasks (Fig. 3 and Tab. 10); its per-setting rows report the same test uncorrected, as a breakdown, and carry no significance marks. All other tests are Holm-corrected [24] for multiple comparisons.

Why the aggregate tests are sign-flip tests on a macro-average. The comparisons between methods (Tabs. 3 and 7 to 9) are not paired Wilcoxon signed-rank tests, which is the more usual choice in this literature: SynthSeg+, from the lineage we compare against, marks a method as better than its competitors using a one-sided Bonferroni-corrected Wilcoxon signed-rank test on paired Dice [4]. We therefore give our reason rather than leave it implicit. The *estimand* we test, by contrast, is not unusual at all — it is the same equal-weight-per-stratum principle the Medical Segmentation Decathlon adopted, for the same reason, as set out below.

The estimand. What we claim is $\text{macro}\Delta$: the mean, over test contrasts, of that contrast’s mean paired per-case difference (ours minus competitor) — equal weight per contrast, which is exactly what every task column of Tab. 3 and every setting column of Tab. 7 reports, so the test and the table measure the same thing. Equal weighting is not cosmetic here. The claim under test is cross-contrast generalization, and a contrast’s held-out case count reflects how large a public cohort happens to be, not how much that contrast should say about generalizing. Abdomen is the extreme case: its CT column pools 140 cases (20 from CHAOS, 100 from AMOS, 20 from SLIVER07), while CHAOS’s three MR contrasts share the same 4 patients and AMOS T2w has 20. Weighting by case count would let CT cast most of the votes in a test whose conclusion is about transfer across contrasts.

This is the same problem, and the same answer, as the Medical Segmentation Decathlon’s. Facing sample sizes that “varied heavily across all tasks and target ROIs,” it ranked algorithms within each target ROI, averaged those ranks per task, and averaged the per-task ranks for the final score — choosing that over a flat average across ROIs precisely because the flat scheme means “tasks with multiple target ROIs ... would be over-weighted” [1]. Our $\text{macro}\Delta$ applies the same principle one level across, with the held-out contrast in place of the target ROI. Where we differ is only in how that weighted quantity is reached: the Decathlon arrives at it through counts of pairwise Wilcoxon

outcomes at an uncorrected per-comparison α , whereas we test the weighted difference itself and Holm-correct across competitors.

Why not a pooled Wilcoxon. The signed-rank test is defined on a single undifferentiated sample of paired differences. Applying it to the concatenation of all contrasts therefore tests a case-weighted null — the imbalance above, reintroduced — and there is no standard reweighting of the signed-rank statistic that recovers the equal-per-contrast estimand. The disagreement this causes runs in both directions: a pooled test can declare a tenth of a Dice point significant because one large cohort supplies most of the cases, or miss a consistent effect confined to the smaller contrasts.

The test. Under the null that the two methods are exchangeable on a given patient, that patient’s paired differences are equally likely to carry either sign. Sign-flipping is therefore a valid randomization null for *any* statistic computed from those differences [12], macro Δ included, which is what makes it applicable where the signed-rank test is not. We report the one-sided p in the direction of the stated hypothesis.

Evaluating the tail. The statistic is a weighted sum of per-patient contributions, so its sign-flip null distribution can be computed rather than sampled. We enumerate all 2^M sign patterns of the $M=20$ patients with the largest contributions exactly, and treat the remaining, lighter patients as a Gaussian term with their exact null variance; with twenty patients or fewer the test is therefore exact. A plain normal approximation to the whole null, which an earlier version of this analysis used, is badly conservative in exactly the designs we have, where a few patients spanning several contrasts carry most of the null variance: on the Abdomen table it returned $p = 0.013$ for a comparison that a 200,000-draw Monte-Carlo test places below 5×10^{-6} . The enumerated tail is deterministic and reaches far tails without the $1/B$ floor of resampling. In the cross-task tests of Tab. 3 (1,456–1,730 patients), the Gaussian remainder carries 7–24% of the null variance for Dice (13–40% for HD95); the exactly enumerated twenty patients carry the rest. Exact enumeration also sets a hard floor for small cohorts: eight patients admit $2^8 = 256$ sign patterns, so no one-sided p can fall below $1/256$ however consistent the effect.

Contrast-group pooling. For Abdomen, the same tested modality is sometimes evidenced by more than one source: Abdomen’s CT evidence comes from CHAOS’s own held-out CT split plus AMOS and SLIVER07. Averaging every raw column with equal weight, the natural first implementation, silently lets a modality with more contributing sources outvote one with fewer in both the task-level average and the significance test, since it then supplies more of the equally-weighted columns being averaged. We instead pool disjoint sources of the *same* modality by concatenat-

ing their cases into one group before any averaging happens, then average modality groups with equal weight. For Abdomen this additionally means averaging equal-weight across liver/spleen/kidney *within* the pooled CT evidence, since AMOS’s own CT column already blends all three organs into one number before it reaches this step. The same pooling applies to Mandible (HaN-Seg and PDDCA both supply CT) and Breast (I-SPY2, Duke and I-SPY1 supply T1-CE, and Duke and I-SPY1 also supply pre-contrast); the remaining tasks have no contrast evidenced by more than one source.

10. Per-setting results

Table 3 in the main paper reports results at the task level, pooling each task’s test contrasts (the training contrast included), all of its training modalities and its external cohorts. This section breaks the same evaluation out into the eighteen settings. Every number is a cross-fold, cross-label average over all test contrasts of the setting, the training contrast included, under the protocol of Sec. 4.1.

Tables 7 and 8 break each task column of Tab. 3 into its training modalities. Where a margin is not significant in a single setting, the honest reading is that one setting cannot separate the methods, not that they are equivalent: several settings have few test units (eight patients on MS, eight sessions on Brain), which is why the main claims rest on the task- and cross-task-level tests.

Boundary distance at task level. Table 9 gives the HD95 counterpart of Tab. 3. Across tasks PALETTE-Aug is significantly better than every competitor, Auglab included. Per task it has the lowest HD95 on Glioma, Abdomen, Brain, MS, Spine and Pelvis (on Abdomen the gap to SRCSM is not significant); it ties Auglab on Breast without a significant gap; and on Mandible it is significantly behind Auglab and nnU-Net, the one task where nnU-Net beats PALETTE-Aug. PALETTE alone, shown for reference, has the lowest HD95 on Breast. When HD95 disagrees with Dice for a method (a noticeably better HD95 than its Dice would predict), the usual explanation is low recall: a method that predicts only a small, confident, well-localized fragment of the structure can have a deceptively good boundary distance while missing most of the target, so a strong HD95 paired with a weak Dice is a warning sign rather than evidence of competitiveness.

Setting	nnU-Net	SynthSeg -noEM	SynthSeg -EM	SRCSM	Auglab	PALETTE- Aug
Glioma T1n	25.3*	8.7*	43.1*	42.7*	46.3	46.4
Glioma T1c	28.3*	9.5*	42.9*	40.2*	46.6	47.1
Glioma T2w	18.1*	8.4*	44.1*	30.1*	47.1	47.3
Glioma FLAIR	21.8*	8.7*	42.3	34.7*	39.4*	43.0
MS FLAIR	25.2*	4.6*	32.8*	27.9*	32.8*	39.5
MS T1w	16.7*	4.0*	35.1*	27.2*	35.8	37.4
Breast T1-CE	34.1*	16.6*	40.4*	16.9*	41.1*	41.9
Breast T2w	35.9*	16.5*	38.9*	12.5*	39.0*	40.5
Abdomen T1in	29.5*	63.3*	84.4*	84.9	84.7*	86.0
Abdomen T2spir	29.6*	58.5*	83.5*	86.3	84.5*	85.8
Brain T1w	26.4*	66.1*	67.0*	66.2	67.1*	67.9
Brain T2w	35.9*	66.6	67.7	66.6	67.2*	67.6
Brain DWI	41.2*	59.7*	63.8	66.6	63.6	64.7
Mandible CBCT	59.2*	53.6*	76.3*	79.8	78.3	77.7
Spine CT	27.0*	46.7*	79.1*	—	77.5*	80.2
Spine Dixon in-phase	69.9*	48.4*	82.2*	—	84.2	84.3
Pelvis CT	35.6*	17.1*	66.5*	67.3	68.5	69.2
Pelvis MRI	24.0*	12.8*	58.6*	58.5*	60.0	62.0

Table 7. Per-setting Dice (%), best per row in bold. Each row is one setting (one training modality of one task), scored on every test contrast of its task, the training contrast and external cohorts included, exactly as its task column of Tab. 3, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

Setting	nnU-Net	SynthSeg -noEM	SynthSeg -EM	SRCSM	Auglab	PALETTE- Aug
Glioma T1n	41.0*	78.0*	18.4*	17.0*	15.1	15.4
Glioma T1c	40.3*	78.3*	19.5*	20.3*	15.1	15.4
Glioma T2w	54.9*	79.7*	16.5*	29.2*	14.8	15.0
Glioma FLAIR	43.1*	76.0*	17.3	20.3*	19.2*	16.6
MS FLAIR	39.9*	41.5*	23.6	23.0	26.6	24.3
MS T1w	30.5*	42.3*	25.9	27.8*	26.3*	24.2
Breast T1-CE	61.8*	103.2*	61.7*	92.7*	57.3	59.0
Breast T2w	55.2	99.1*	60.1*	92.1*	56.9	56.3
Abdomen T1in	119.0*	78.0*	25.4	29.4*	30.4*	22.3
Abdomen T2spir	125.1*	83.2*	22.0	21.7	25.1	23.2
Brain T1w	44.3*	11.2*	7.2	6.6	7.4	6.9
Brain T2w	27.8*	9.5*	4.0	4.6*	4.3*	4.1
Brain DWI	18.8*	5.7*	4.7	4.7*	4.7	4.6
Mandible CBCT	21.7	32.9*	24.9*	21.9	22.1	23.5
Spine CT	78.3*	110.4*	4.5*	—	5.8*	3.9
Spine Dixon in-phase	21.6*	113.2*	3.5*	—	3.6	3.1
Pelvis CT	146.2*	252.0*	20.0	22.3	23.5	21.6
Pelvis MRI	152.3*	271.6*	48.0	56.9*	47.8*	36.4

Table 8. Per-setting HD95 (mm), best per row in bold. Each row is one setting (one training modality of one task), scored on every test contrast of its task, the training contrast and external cohorts included, exactly as its task column of Tab. 3, so the rows of a task average to that column. *: PALETTE-Aug significantly better than that method within the setting (same patient-level sign-flip test, Holm-corrected across competitors). Spine has no SRCSM run.

	Glioma	MS	Breast	Abdomen	Brain	Mandible	Spine	Pelvis	p vs ours
nnU-Net	44.8	35.2	58.5	122.1	30.3	21.7	49.9	149.3	2×10^{-47}
SynthSeg-noEM	78.0	41.9	101.1	80.6	8.8	32.9	111.8	261.8	2×10^{-58}
SynthSeg-EM	17.9	24.7	60.9	23.7	5.3	24.9	4.0	34.0	2.7×10^{-6}
SRCSM	21.7	25.4	92.4	25.6	5.3	21.9	—	39.6	1×10^{-43}
Auglab	16.1	26.5	57.1	27.8	5.4	22.1	4.7	35.6	2×10^{-4}
PALETTE alone (ours)	16.6	23.3	56.3	24.9	4.9	22.3	—	35.2	$0.23^{\S}$
PALETTE-Aug (ours)	15.6	24.3	57.7	22.7	5.2	23.5	3.5	29.0	—
competitors beaten ($p < 0.05$)	5/5	3/5	3/5	4/5	2/5	2/5	4/4	5/5	

Table 9. Task-level HD95 (mm, $\downarrow$). Same pooling and test as Tab. 3, and likewise no cross-task average; the bottom row counts the competitors PALETTE-Aug beats within each task (on Mandible, Auglab and nnU-Net are significantly ahead of it). Spine has no SRCSM run and no ladder. PALETTE alone: the transform without Auglab, left out of the bolding; § PALETTE-Aug against PALETTE alone over the seven tasks with both. SRCSM includes its test-time source matching.

11. Ablation-ladder details

Tables 10 and 11 give the fill swap of every setting, Fig. 5 its HD95 counterpart per task, Tab. 12 every rung, and Tab. 13 every training $\rightarrow$ test contrast pair.

Full ladder: numbers and significance. Figure 3 (main paper) shows the rung sequence visually; Tab. 12 gives the underlying out-of-domain Dice at every rung of all sixteen ladders, with significance for each step against the rung immediately before it (Holm-corrected paired Wilcoxon on patients, within each ladder’s family of steps). Most steps that add an ingredient help, as expected. The significant *decreases* (marked $\downarrow$) are the Voronoi step on Glioma T1n and both Abdomen settings, which the real-fill step then more than recovers; the fill swap on Brain T1w and T2w; and the +Auglab step on MS T1w, both Abdomen settings, Brain DWI and Mandible, the task where the complete method places third (Sec. 4.2), plus Mandible’s step to synthetic-contrast validation. On Mandible, the noise-fill rung (69.8) is the ladder’s maximum: the fill swap does not change Dice significantly and adding the Auglab recipe takes it further down; adding PALETTE to Auglab also lowers Mandible Dice (Tab. 3), so the two components interact negatively there and we cannot attribute the shortfall to either; we have not diagnosed which Auglab operator is responsible. Abdomen T1in, MS T1w and Brain DWI also peak at the real-fill rung, while on all four Glioma settings and on Breast and Pelvis the +Auglab step adds Dice.

Voronoi’s own marginal contribution is task-dependent rather than uniformly positive. A separate paired significance check on all-contrast Dice (the estimand of Tab. 17, not the OOD slice above) shows that Voronoi significantly *helps* MS FLAIR and (by a small margin) Brain T1w, and significantly *hurts* Glioma T1n and, by a small margin, Abdomen T1in ($87.7 \rightarrow 86.9$, $p = 2 \times 10^{-4}$, 24 patients; its OOD step above is a small significant decrease too). The full method is still best on Glioma T1n (Tab. 7; not significantly ahead of Auglab): whatever Voronoi costs there is more than recovered by the subsequent real-intensity fill step, so the net effect of the complete pipeline is positive even though one internal component is not.

Table 13 breaks the fill swap down by training $\rightarrow$ test contrast pair: the numbers behind the thin curves of Fig. 3.

Per-label breakdown of the fill swap. Glioma T2w’s pooled fill-swap effect is small and not significant (Tab. 10), with one significantly negative held-out contrast inside it (T2w $\rightarrow$ T1n, Tab. 13). This raises a further question: within that one (training, eval) pair, does the fill-swap step affect every anatomical sub-region equally, or is the worsening itself concentrated in specific labels? We break Glioma T2w $\rightarrow$ T1n down by its four labels: necrotic core (NCR),

Task / setting	noise	Δ	rel. (%)	p
<i>Interface-bounded</i>				
Abdomen (all settings)	82.9	+3.51*	+4.2	1.2e-23
Abdomen T1in	85.3	+1.77	+2.1	3e-25
Abdomen T2spir	80.5	+5.25	+6.5	2.3e-11
Brain (all settings)	TODO	TODO	TODO	TODO
Brain T1w	TODO	TODO	TODO	TODO
Brain T2w	TODO	TODO	TODO	TODO
Brain DWI	TODO	TODO	TODO	TODO
Mandible (all settings)	69.8	−1.56	−2.2	0.52
Mandible CBCT [‡]	69.8	−1.56	−2.2	0.52
Pelvis (all settings)	49.6	+3.12*	+6.3	0.0037
Pelvis CT	34.1	+5.86	+17.2	0.0014
Pelvis MRI	65.0	+0.67	+1.0	0.12
<i>Appearance-defined</i>				
Glioma (all settings)	TODO	TODO	TODO	TODO
Glioma T1n	TODO	TODO	TODO	TODO
Glioma T1c	TODO	TODO	TODO	TODO
Glioma T2w	TODO	TODO	TODO	TODO
Glioma FLAIR	TODO	TODO	TODO	TODO
MS (all settings)	TODO	TODO	TODO	TODO
MS FLAIR	TODO	TODO	TODO	TODO
MS T1w	TODO	TODO	TODO	TODO
Breast (all settings)	34.3	+1.53*	+4.5	2.2e-16
Breast T1-CE	29.2	+0.41	+1.4	0.0032
Breast T2w	39.3	+2.65	+6.8	1.1e-20

Table 10. Replacing SynthSeg-style noise fill with the real-intensity remap, *with the partition and region target means unchanged*. noise: out-of-domain Dice with noise fill; Δ : Dice change at the swap; rel.: Δ relative to the noise-fill Dice. Significance is tested once per task (bold rows; the test annotated in Fig. 3): two-sided paired Wilcoxon on patients, each patient contributing its mean difference over the held-out contrasts and training modalities it appears in, Holm-corrected across the seven tasks (*: $p < 0.05$). The indented setting rows break each task into its training modalities; their p is the same test within the setting, uncorrected and without significance marks, since the claim is made at the task level. With eight test patients (MS) or eight sessions of four subjects (Brain) a two-sided p cannot fall below $1/128$. [‡]Held-out contrasts are cross-dataset (HaN-Seg, PDDCA), since ToothFairy2 is CBCT only. Per pair: Tab. 13; HD95: Tab. 11.

surrounding non-enhancing FLAIR hyperintensity (SNFH), enhancing tumor (ET) and resection cavity (RC); $n = 70$ test cases for SNFH and ET, present in essentially every case (NCR $n = 51$, RC $n = 67$, since not every tumor has a necrotic core or a resection cavity).

The worsening is concentrated in SNFH (Tabs. 14 and 15). SNFH has the largest gap between directions (+23.2 Dice) and the largest significant sign reversal: +14.8 when trained on T1n, −11.9 when trained on T2w. RC reverses too, at a smaller size. NCR and ET show no significant effect in either direction. Figure 6 shows this on one case: with real fill, the T2w-trained model’s SNFH pre-

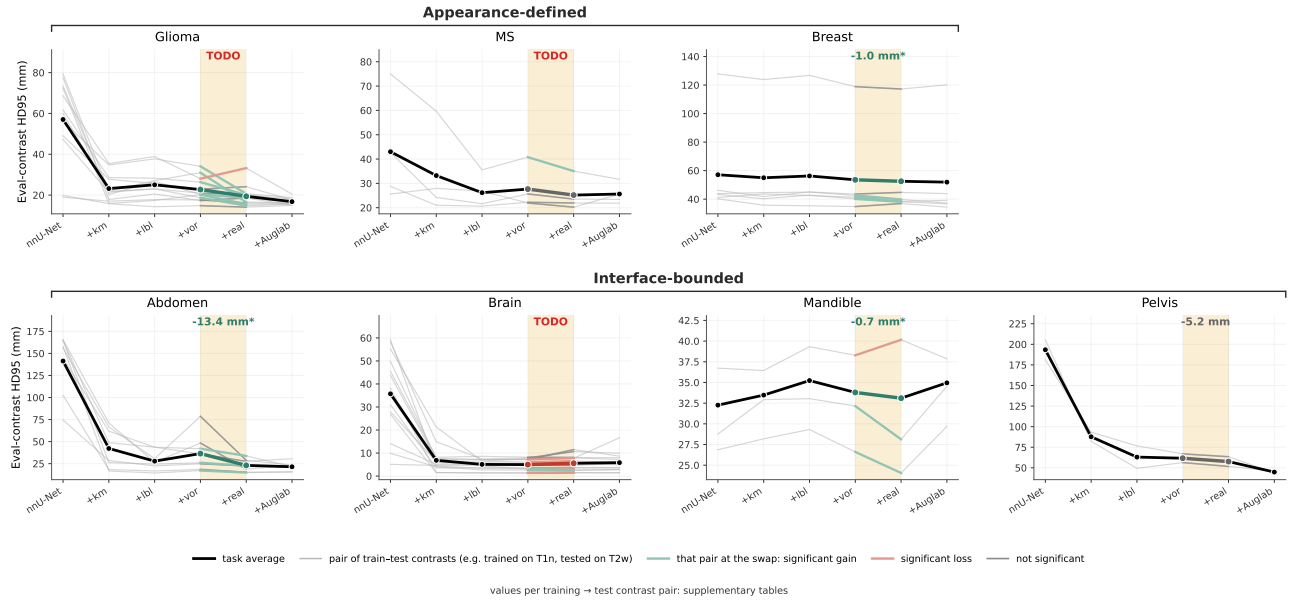


Figure 5. Figure 3 (main paper) on HD95 instead of Dice: same tasks, grouping and visual scheme. Annotation: the task’s HD95 change at the fill swap in mm (negative is better); *: significant. Glioma, Abdomen, Breast and Mandible improve significantly, Brain worsens significantly, and MS and Pelvis do not move significantly. Per-setting numbers: Tab. 11.

1299 diction almost vanishes (label Dice 0.90 $\rightarrow$ 0.06), while its
1300 ET prediction is almost unchanged (0.61 $\rightarrow$ 0.68).

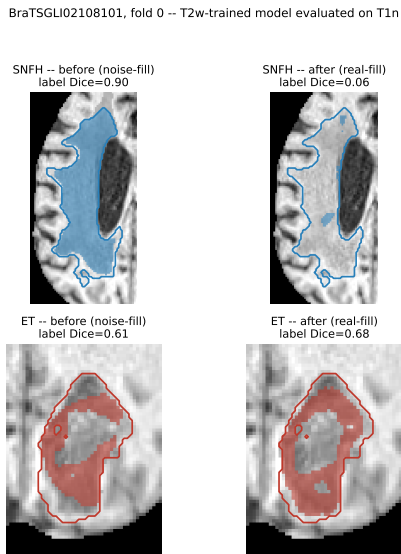


Figure 6. One Glioma test case, fold 0, T2w-trained model evaluated on T1n. Zoomed crops before (noise-fill) and after (real-fill) the fill-swap step, for the SNFH-dominant (top) and ET-dominant (bottom) lesion components of the same case. Color = ground truth wherever the prediction agrees; gray = false positive; outline = ground truth.

One candidate mechanism, flagged for future work rather than tested here: a texture pattern learned from one contrast’s rendering of a diffuse, texture-defined label like edema may not simply fail to transfer to a different contrast’s rendering of the same tissue; it may actively mislead, if the model has come to rely on a contrast-specific cue that the other contrast does not reproduce. This would be an instance of shortcut learning [16]: a network relying on a domain-specific cue that does not reproduce across acquisition protocols, the mechanism Ouyang et al. [45] identify behind failed single-source domain generalization in cross-modality and cross-sequence medical segmentation, rather than on the intended, causal feature, which a simplicity-biased network need not prefer even when it would generalize better [52]. We have not tested this mechanistically here; it is the most concrete hypothesis consistent with the label-specific, direction-dependent pattern above, not a demonstrated cause.

Task / setting	noise	Δ (mm)	rel. (%)	p
<i>Interface-bounded</i>				
Abdomen (all settings)	36.4	-13.38*	-36.7	2.4e-9
Abdomen T1in	26.0	-4.06	-15.6	1.5e-13
Abdomen T2spir	46.9	-22.70	-48.4	3e-5
Brain (all settings)	TODO	TODO	TODO	TODO
Brain T1w	TODO	TODO	TODO	TODO
Brain T2w	TODO	TODO	TODO	TODO
Brain DWI	TODO	TODO	TODO	TODO
Mandible (all settings)	33.8	-0.69*	-2.1	0.031
Mandible CBCT [†]	33.8	-0.69	-2.1	0.0093
Pelvis (all settings)	61.6	-5.21	-8.5	0.47
Pelvis CT	67.0	-3.48	-5.2	0.19
Pelvis MRI	56.3	-4.62	-8.2	0.75
<i>Appearance-defined</i>				
Glioma (all settings)	TODO	TODO	TODO	TODO
Glioma T1n	TODO	TODO	TODO	TODO
Glioma T1c	TODO	TODO	TODO	TODO
Glioma T2w	TODO	TODO	TODO	TODO
Glioma FLAIR	TODO	TODO	TODO	TODO
MS (all settings)	TODO	TODO	TODO	TODO
MS FLAIR	TODO	TODO	TODO	TODO
MS T1w	TODO	TODO	TODO	TODO
Breast (all settings)	53.6	-1.03*	-1.9	0.016
Breast T1-CE	67.5	-0.83	-1.2	0.044
Breast T2w	39.6	-1.23	-3.1	8.2e-5

Table 11. Same fill-swap step as Tab. 10, on HD95 (negative is better; noise: HD95 with noise fill), with the same tests: task rows Holm-corrected across the seven tasks (*: $p < 0.05$), setting rows uncorrected. HD95 is less consistent than Dice within a task (on Glioma it improves from three training contrasts and worsens from FLAIR), so we state the claim on Dice and report HD95 here rather than as confirmation of it.

Rung	Glioma T1n	Glioma T1c	Glioma T2w	Glioma FLAIR	MS FLAIR	MS T1w	Breast T1-CE	Breast T2w
nnU-Net	16.5	15.9	6.0	11.0	2.4	1.1	23.7	33.5
+k-means	38.4***	29.6***	38.6***	26.1***	11.2*	9.8*	28.8***	37.0***
+label remap	41.8***	30.3	40.2**	26.4	17.5*	22.4*	28.5	38.8***
+Voronoi	TODO	TODO	TODO	TODO	TODO	TODO	29.2	39.3
real fill	TODO	TODO	TODO	TODO	TODO	TODO	29.6**	41.9***
+Auglab	46.9	44.2	46.8***	40.9***	30.6	36.4*↓	34.7***	41.9
+Auglab, s.v.	47.6*	46.5***	46.9	41.2	31.3	34.7	—	—

Rung	Abdomen T1in	Abdomen T2spir	Brain T1w	Brain T2w	Brain DWI	Mandible CBCT	Pelvis CT	Pelvis MRI
nnU-Net	17.0	19.7	13.6	26.4	34.6	39.4	0.7	10.0
+k-means	82.2***	77.4***	62.2*	63.3*	60.5*	64.6***	22.6***	55.9***
+label remap	86.1***	81.4***	62.9	63.2	63.6*	65.6***	30.0***	63.3***
+Voronoi	85.3***↓	80.5***↓	TODO	TODO	TODO	69.8***	34.1*	65.0**
real fill	87.1***	85.7***	TODO	TODO	TODO	68.2	40.0**	65.7
+Auglab	86.4***↓	85.4***↓	63.3	63.7	62.7*↓	67.6***↓	46.3**	67.2
+Auglab, s.v.	86.1	85.9	63.6*	64.1	61.5	67.5***↓	—	—

Table 12. Out-of-domain Dice (%) at every rung of every ladder (Fig. 3), with significance for each step against the rung above it (“s.v.”, synthetic-contrast validation, against real-contrast validation): * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (Holm-corrected paired Wilcoxon on patients, within each ladder; Tab. 10 tests the fill swap once per task instead); ↓ marks a significant *decrease*. Rungs: nnU-Net, +k-means, +label remap, +Voronoi (noise fill), real fill (PALETTE alone), +Auglab (PALETTE-Aug); Fig. 3 abbreviates them nnU-Net, +km, +lbl, +vor, +real, +Auglab. MS and Brain have eight test units each, so their smallest attainable p after correction is just under 0.05. Breast and Pelvis have no synthetic-validation arm. Mandible’s held-out contrasts all come from external cohorts.

Task	Train → test	Noise	Real	Δ	Rel. (%)	p
<i>Appearance-defined</i>						
Glioma	T1n → T1c	TODO	TODO	TODO	TODO	TODO
	T1n → T2w	TODO	TODO	TODO	TODO	TODO
	T1n → FLAIR	TODO	TODO	TODO	TODO	TODO
	T1c → T1n	TODO	TODO	TODO	TODO	TODO
	T1c → T2w	TODO	TODO	TODO	TODO	TODO
	T1c → FLAIR	TODO	TODO	TODO	TODO	TODO
	T2w → T1n	TODO	TODO	TODO	TODO	TODO
	T2w → T1c	TODO	TODO	TODO	TODO	TODO
	T2w → FLAIR	TODO	TODO	TODO	TODO	TODO
	FLAIR → T1n	TODO	TODO	TODO	TODO	TODO
	FLAIR → T1c	TODO	TODO	TODO	TODO	TODO
	FLAIR → T2w	TODO	TODO	TODO	TODO	TODO
	MS	TODO	TODO	TODO	TODO	TODO
	FLAIR → T1w	TODO	TODO	TODO	TODO	TODO
	FLAIR → T2w	TODO	TODO	TODO	TODO	TODO
	T1w → FLAIR	TODO	TODO	TODO	TODO	TODO
	T1w → T2w	TODO	TODO	TODO	TODO	TODO
Breast	T1-CE → T2w	22.7	23.1	+0.5	+2.0	0.08
	T1-CE → pre-contrast	32.4	31.0	-1.4	-4.4	0.10
	T1-CE → DWI	32.6	34.8	+2.2*	+6.8	8×10^{-9}
	T2w → T1-CE	50.3	52.9	+2.6*	+5.2	4×10^{-8}
	T2w → pre-contrast	40.1	40.5	+0.4	+0.9	0.09
	T2w → DWI	27.5	32.5	+5.0*	+18.2	2×10^{-18}
<i>Interface-bounded</i>						
Abdomen	T1in → T1out	85.8	86.1	+0.3	+0.4	0.50
	T1in → T2spir	86.2	87.7	+1.6	+1.8	0.50
	T1in → CT	82.0	85.2	+3.2*	+3.9	3×10^{-21}
	T1in → T2w	87.4	89.4	+2.0*	+2.3	2×10^{-4}
	T2spir → T1in	74.6	83.6	+8.9	+12.0	0.25
	T2spir → T1out	80.3	87.7	+7.3	+9.1	0.25
	T2spir → CT	78.6	81.5	+2.9*	+3.7	1×10^{-8}
	T2spir → T2w	88.4	90.2	+1.8*	+2.0	0.02
Brain	T1w → T2w	TODO	TODO	TODO	TODO	TODO
	T1w → BOLD	TODO	TODO	TODO	TODO	TODO
	T1w → DWI	TODO	TODO	TODO	TODO	TODO
	T1w → EPI	TODO	TODO	TODO	TODO	TODO
	T1w → GRE	TODO	TODO	TODO	TODO	TODO
	T2w → T1w	TODO	TODO	TODO	TODO	TODO
	T2w → BOLD	TODO	TODO	TODO	TODO	TODO
	T2w → DWI	TODO	TODO	TODO	TODO	TODO
	T2w → EPI	TODO	TODO	TODO	TODO	TODO
	T2w → GRE	TODO	TODO	TODO	TODO	TODO
	DWI → T1w	TODO	TODO	TODO	TODO	TODO
	DWI → T2w	TODO	TODO	TODO	TODO	TODO
	DWI → BOLD	TODO	TODO	TODO	TODO	TODO
	DWI → EPI	TODO	TODO	TODO	TODO	TODO
	DWI → GRE	TODO	TODO	TODO	TODO	TODO
Mandible	CBCT → HaN-Seg CT	83.8	83.7	-0.1	-0.1	0.39
	CBCT → HaN-Seg MR-T1	57.2	53.3	-3.9*	-6.8	9×10^{-9}
	CBCT → PDDCA CT	80.8	82.4	+1.7*	+2.1	2×10^{-5}
Pelvis	CT → MRI	34.1	40.0	+5.9*	+17.2	1×10^{-3}
	MRI → CT	65.0	65.7	+0.7	+1.0	0.12

Table 13. The fill swap for each training → test contrast pair: out-of-domain Dice (%) of the noise-fill rung and of the real-fill rung (PALETTE alone), their difference, the difference relative to the noise-fill Dice, and the pair’s paired Wilcoxon test on patients, Holm-corrected across the held-out contrasts of its setting (*: $p < 0.05$, the pairs colored in Fig. 3; Tab. 10 tests each setting as a whole). External cohorts are pooled with the training cohort by contrast, except Mandible, whose held-out contrasts all come from external cohorts and are listed per cohort. 50 pairs.

Label	T1n→T2w	T2w→T1n	gap
NCR	18.3	21.5	-3.2
SNFH	66.4	43.2	+23.2
ET	27.2	26.9	+0.4
RC	50.6	43.1	+7.5

Table 14. Absolute OOD Dice (%) of the real-fill rung (PALETTE alone), per label, both transfer directions.

Label	T1n→T2w Δ	T2w→T1n Δ
NCR	+0.4 ($p = 0.25$)	+0.8 ($p = 1.0$)
SNFH	+14.8* ($p = 5.4 \times 10^{-12}$)	-11.9* ($p = 1.6 \times 10^{-7}$)
ET	+2.0 ($p = 0.054$)	-0.2 ($p = 1.0$)
RC	+3.5* ($p = 6 \times 10^{-5}$)	-6.5* ($p = 0.005$)

Table 15. Fill-swap step (+Voronoi noise fill → real fill), per label, both directions. p : Holm-corrected paired Wilcoxon, family = the four labels within each direction; *: significant.

12. Synthetic contrast at validation time

Throughout the main paper PALETTE-Aug is applied with synthetic contrast during training only, and the best checkpoint is chosen on real-contrast validation images (*val000*), as for every competing method and every ladder rung; this keeps the comparison like-for-like. PALETTE-Aug can additionally be applied to the validation set, so that model selection is performed under randomized contrast as well (*val100*).

Validation-time randomization improves Glioma and Abdomen on both metrics (Tab. 16), most visibly on boundary distance for Abdomen (22.7 $\rightarrow$ 20.6 mm). Across tasks the HD95 gain is significant but the Dice change is not, and MS and Brain Dice move slightly the other way, so we do not claim it as a separate contribution; we report it because it costs nothing and because it indicates that the benefit of contrast randomization extends to model selection and is not confined to the gradient updates.

Task	Dice $\uparrow$		HD95 (mm) $\downarrow$	
	val000	val100	val000	val100
Glioma	45.9	46.6	15.6	15.1
Abdomen	85.9	86.0	22.7	20.6
Brain	66.7	66.5	5.2	4.7
MS	38.5	37.9	24.3	24.2

Table 16. Applying PALETTE-Aug at validation time as well as at training time, on the four tasks with a task-level table for both configurations (Mandible’s val100 arm is scored only in its ladder, Tab. 12; the Breast, Spine and Pelvis arms were not trained). Across these tasks the two do not differ on Dice (same cross-task test as Tab. 3, two-sided $p = 0.93$), and val100 has a lower HD95 (two-sided $p = 0.031$, -0.8 mm). We report the like-for-like val000 configuration throughout the main paper.

13. Component ablation on all test contrasts

To attribute the gain of the full method to its individual components, rather than to synthetic-contrast augmentation in general, we build an additive ablation ladder on top of nnU-Net with its default augmentation (Sec. 4.1): first the unsupervised k -means intensity partition alone (§3.1, step 1); then the anatomical label-remap step (step 2); then the Voronoi spatial sub-parcellation (step 3); and finally the real fill through the signed affine remap of Eq. (1) (step 4), which together constitute PALETTE alone. Every rung is evaluated without Auglab, so that the ladder isolates PALETTE’s own internal components from the separate question of whether it composes with a prior augmentation pipeline, which we address next. The synthetic-data fraction is fixed at 50% of training iterations throughout, as one of several augmentation constants (alongside, e.g., the k -means class count and the affine magnitude range of §3.1) that are set once and not tuned per dataset; validation uses real-contrast data only (no synthetic contrast at validation time).

Composability with Auglab. PALETTE-Aug is deployed, throughout this paper, as one additional operator inside the Auglab pipeline rather than as a replacement for it (Sec. 4.1); Auglab is a full prior augmentation method in its own right, not a component of PALETTE-Aug, so stacking it on top of the ladder above is a composability/deployment check, not a further component-ablation step. Table 17 keeps the two separate: the top five rows isolate PALETTE’s internal components, evaluated with no Auglab throughout, while the bottom row reports PALETTE-Aug plus Auglab — the configuration used everywhere else in this paper — set apart by a rule rather than presented as the next rung of the same ladder.

Component	Abdomen T1in	MS FLAIR	Glioma T1n	Brain T1w
nnU-Net (default augmentation)	36.1	25.2	26.0	26.4
+ k -means partition	84.8	30.6	41.7	67.4
+ label remap	87.7	34.6	44.3	68.0
+ Voronoi sub-parcellation	86.9	TODO	TODO	TODO
real fill (PALETTE alone)	87.9	40.4	47.6	66.9
+ Auglab (PALETTE-Aug)	87.4	39.5	47.5	67.9

Table 17. Component ablation (top five rows, additive, no Auglab) and composability check (bottom row). From the k -means row on, each rung adds one PALETTE-Aug component on top of the row above; the first step also replaces nnU-Net’s default intensity augmentation with spatial augmentation only. Dice (%) over all test contrasts of the setting, the training contrast included, with equal weight per contrast (the all-contrast series of the same ladders as Tab. 12, which reports the held-out contrasts only). Abdomen T1in: CHAOS’s own test contrasts only, without the external cohorts. The nnU-Net and PALETTE-Aug rows are the runs of Tab. 7; their values can differ slightly from it (Glioma), because the ladder averages over the labels present in each case before averaging cases, whereas Tab. 7 averages each label over cases first.

14. Texture fidelity per method

The per-method NGF analysis is in the main paper (Fig. 4); Tab. 18 gives its per-task values. Figure 7 follows NGF and Dice along the ladder rungs on MS FLAIR (an earlier, FLAIR-only NGF run on 30 scans): the three noise-fill rungs stay at NGF 0.45–0.46 while their Dice rises from 11.2 to 19.6, through contrast randomization, not texture; only the real-fill swap raises both. (Table 18 averages FLAIR and T1w over 20 scans each, hence 0.42 for MS noise fill there.)

Auglab keeps the most texture among the competitors. Auglab keeps less source texture than PALETTE alone on six of seven tasks but far more than the label-style methods (Tab. 18), which fits its standing as the best competitor on five tasks (Tab. 3). The large margins appear against the methods that discard texture.

Method	Glioma	MS	Breast	Abdomen	Brain	Mandible	Pelvis	Mean
PALETTE alone	0.836	0.842	0.809	0.805	0.721	0.904	0.863	0.826
PALETTE-Aug	0.722	0.713	0.693	0.701	0.681	0.718	0.705	0.705
Auglab	0.757	0.758	0.728	0.735	0.727	0.730	0.733	0.738
SRCSM	0.663	0.658	0.645	0.636	0.626	0.600	0.662	0.641
SynthSeg-EM	0.540	0.454	0.494	0.535	0.535	0.465	0.469	0.499
SynthSeg-noEM	0.337	0.337	0.332	0.374	0.504	0.360	0.339	0.369
noise fill	0.465	0.423	0.448	0.463	0.463	0.373	0.416	0.436

Table 18. Texture fidelity per task: NGF similarity of the augmented volume to its source scan (1 = all texture kept, $\approx 1/3$ = isotropic noise) on the seven tasks with a ladder (Spine has none), mean over contrasts of the mean over scans (16–20 per contrast, 5 draws each). PALETTE alone is the ladder’s real-fill rung. PALETTE-Aug applies PALETTE in half of its draws, first in the Auglab chain, whose operators then act on the remapped image: the combination keeps less texture than either PALETTE alone or Auglab. PALETTE alone keeps more texture than every other method on every task, except Auglab on Brain ($\Delta = -0.005$, $p = 0.68$), where the difference is not significant (paired one-sided Wilcoxon over scans, Holm-corrected over the 42 comparisons; largest significant $p = 4.1 \times 10^{-3}$).

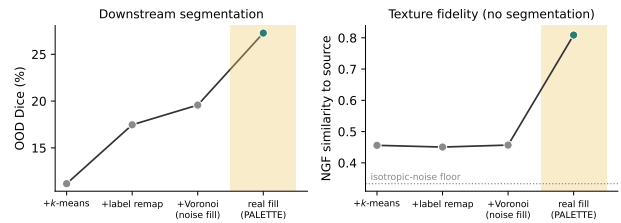


Figure 7. Noise versus real fill on MS FLAIR. Left: out-of-domain Dice at each rung. Right: NGF similarity of the augmented volume to its source (mean $\pm$ s.e. over 30 scans). The noise-fill rungs keep only the partition edges while Dice rises; the real-fill swap (shaded) lifts both.